

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0879

Adabell 100 kV
October 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0879. The Interconnection Customer proposes a 295.9 MW solar photovoltaic facility interconnecting at the Adabell 100 kV substation in the Illinois area. The request was received on October 15, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 390 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,270,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Omni-Hackberry 500/230kV TX
Contingency: Omni-Hackberry-Norwich 500kV
Loading: 111.2% (Rating: 1,409 MVA)
DFAX: 5.5%
Pre-project loading: 110.0%
2. Facility: Waldo 500-230 kV Auto
Contingency: Trego-Beaver 500kV
Loading: 102.7% (Rating: 716 MVA)
DFAX: 9.8%
Pre-project loading: 98.7%
3. Facility: Hebo-Buris Pit Substation 230kV
Contingency: Beaver-Trego 500kV
Loading: 101.5% (Rating: 777 MVA)
DFAX: 8.0%
Pre-project loading: 98.4%
4. Facility: Truman-Waldo 230kV
Contingency: Beaver-Trego 500kV
Loading: 101.4% (Rating: 546 MVA)
DFAX: 9.9%
Pre-project loading: 96.0%
5. Facility: Burris Pit-James De Young Substation 230kV
Contingency: Trego-Beaver Substation 500kV
Loading: 100.8% (Rating: 801 MVA)
DFAX: 8.0%
Pre-project loading: 97.8%

6. Facility: Station S Van Nuys-Gould 230kV
Contingency: Troy Energy Llc-Richland 500kV
Loading: 100.7% (Rating: 511 MVA)
DFAX: 6.6%
Pre-project loading: 96.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Omni-Hackberry 500/230 kV Auto	Replace transformer with higher-rated unit	\$31,515,000	2% (\$682,000)
Waldo 500/230 kV TX	Replace transformer with higher-rated unit	\$31,960,000	2% (\$736,000)
Burris Pit-Hebo 230kV	Reconductor line	\$7,484,000	2% (\$163,000)
Truan-Waldo 230kV	Reconductor line	\$5,668,000	2% (\$131,000)
Burris Pit-James De Young Substation 230kV	Reconductor line	\$19,761,000	2% (\$431,000)
Station S Van Nuys-Gould 230kV	Reconductor line	\$5,040,000	2% (\$127,000)
	Total allocated to project		\$2,270,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.